EVERY FASTENER PERMUTATION BEHIND THE PLATE

39 combinations: 11 nut constructions x 3 washer options, and with/without threadlocker for the nuts that do not lock on their own.

The governing sum, every row: **plate + washer + nut <= stud**

plate 4.75 mm (0.187 in) - stud 12.70 mm (0.500 in), fixed by the magnet - plate hole O8.5 mm.

Washer thickness is the MAX of the range McMaster sells it to, never the nominal: a stack that must fit has to survive the thickest one shipped.

"bearing" is the annulus the OUTERMOST part presses on the plate with — the washer if there is one, else the nut's own face. Pressure is at the magnet's 61.2 lb.

Nut and washer part numbers are the BLACK-OXIDE ones where stocked; a number in grey means only plain 18-8 exists.

#	nut	nut ht	washer	wshr t	plate + washer + nut	= needs	vs stud	slack	locking
FITS — 11	at least 0.50 mm of thread to spare								
1	THIN nylon-insert locknut	6.35	—	0.00	4.75 + 0.00 + 6.35	11.10	12.70	**+1.60**	**mechanical**
2	distorted-thread locknut (centre-lock)	6.75	—	0.00	4.75 + 0.00 + 6.75	11.50	12.70	**+1.20**	**mechanical**
3	distorted-thread locknut (top-lock)	6.75	—	0.00	4.75 + 0.00 + 6.75	11.50	12.70	**+1.20**	**mechanical**
4	JAM nut (half height)	4.76	—	0.00	4.75 + 0.00 + 4.76	9.51	12.70	**+3.19**	**chemical (threadlocker)**
5	**JAM nut (half height) <<< SPECIFIED**	4.76	OVERSIZED washer O1.250	1.52	4.75 + 1.52 + 4.76	11.04	12.70	**+1.66**	**chemical (threadlocker)**
6	JAM nut (half height)	4.76	flat washer O0.750	1.52	4.75 + 1.52 + 4.76	11.04	12.70	**+1.66**	**chemical (threadlocker)**
7	standard hex nut	6.75	—	0.00	4.75 + 0.00 + 6.75	11.50	12.70	**+1.20**	**chemical (threadlocker)**
8	JAM nut (half height)	4.76	—	0.00	4.75 + 0.00 + 4.76	9.51	12.70	**+3.19**	NONE
9	JAM nut (half height)	4.76	OVERSIZED washer O1.250	1.52	4.75 + 1.52 + 4.76	11.04	12.70	**+1.66**	NONE
10	JAM nut (half height)	4.76	flat washer O0.750	1.52	4.75 + 1.52 + 4.76	11.04	12.70	**+1.66**	NONE
11	standard hex nut	6.75	—	0.00	4.75 + 0.00 + 6.75	11.50	12.70	**+1.20**	NONE
INSIDE THE TOLERANCE STACK — 12	within +/-0.50 mm — three stacked tolerances can swallow this; do not build on it								
12	THIN nylon-insert locknut	6.35	OVERSIZED washer O1.250	1.52	4.75 + 1.52 + 6.35	12.62	12.70	**+0.08**	**mechanical**
13	THIN nylon-insert locknut	6.35	flat washer O0.750	1.52	4.75 + 1.52 + 6.35	12.62	12.70	**+0.08**	**mechanical**
14	THIN-HEAVY nylon-insert locknut	7.94	—	0.00	4.75 + 0.00 + 7.94	12.69	12.70	**+0.01**	**mechanical**
15	distorted-thread locknut (centre-lock)	6.75	OVERSIZED washer O1.250	1.52	4.75 + 1.52 + 6.75	13.02	12.70	**-0.32**	**mechanical**
16	distorted-thread locknut (top-lock)	6.75	OVERSIZED washer O1.250	1.52	4.75 + 1.52 + 6.75	13.02	12.70	**-0.32**	**mechanical**
17	distorted-thread locknut (centre-lock)	6.75	flat washer O0.750	1.52	4.75 + 1.52 + 6.75	13.02	12.70	**-0.32**	**mechanical**
18	distorted-thread locknut (top-lock)	6.75	flat washer O0.750	1.52	4.75 + 1.52 + 6.75	13.02	12.70	**-0.32**	**mechanical**
19	locknut with external-tooth lock washer (keps)	8.33	—	0.00	4.75 + 0.00 + 8.33	13.08	12.70	**-0.38**	**mechanical**
20	standard hex nut	6.75	OVERSIZED washer O1.250	1.52	4.75 + 1.52 + 6.75	13.02	12.70	**-0.32**	**chemical (threadlocker)**
21	standard hex nut	6.75	flat washer O0.750	1.52	4.75 + 1.52 + 6.75	13.02	12.70	**-0.32**	**chemical (threadlocker)**
22	standard hex nut	6.75	OVERSIZED washer O1.250	1.52	4.75 + 1.52 + 6.75	13.02	12.70	**-0.32**	NONE
23	standard hex nut	6.75	flat washer O0.750	1.52	4.75 + 1.52 + 6.75	13.02	12.70	**-0.32**	NONE
DOES NOT FIT — 16	the stud ends before the nut does								
24	nylon-insert FLANGE nut	8.73	—	0.00	4.75 + 0.00 + 8.73	13.48	12.70	**-0.78**	**mechanical**
25	nylon-insert locknut	8.73	—	0.00	4.75 + 0.00 + 8.73	13.48	12.70	**-0.78**	**mechanical**
26	flex-top locknut	9.13	—	0.00	4.75 + 0.00 + 9.13	13.88	12.70	**-1.18**	**mechanical**
27	THIN-HEAVY nylon-insert locknut	7.94	OVERSIZED washer O1.250	1.52	4.75 + 1.52 + 7.94	14.21	12.70	**-1.51**	**mechanical**
28	THIN-HEAVY nylon-insert locknut	7.94	flat washer O0.750	1.52	4.75 + 1.52 + 7.94	14.21	12.70	**-1.51**	**mechanical**
29	locknut with external-tooth lock washer (keps)	8.33	OVERSIZED washer O1.250	1.52	4.75 + 1.52 + 8.33	14.61	12.70	**-1.91**	**mechanical**
30	locknut with external-tooth lock washer (keps)	8.33	flat washer O0.750	1.52	4.75 + 1.52 + 8.33	14.61	12.70	**-1.91**	**mechanical**
31	nylon-insert locknut	8.73	OVERSIZED washer O1.250	1.52	4.75 + 1.52 + 8.73	15.01	12.70	**-2.31**	**mechanical**
32	nylon-insert FLANGE nut	8.73	OVERSIZED washer O1.250	1.52	4.75 + 1.52 + 8.73	15.01	12.70	**-2.31**	**mechanical**
33	nylon-insert locknut	8.73	flat washer O0.750	1.52	4.75 + 1.52 + 8.73	15.01	12.70	**-2.31**	**mechanical**
34	nylon-insert FLANGE nut	8.73	flat washer O0.750	1.52	4.75 + 1.52 + 8.73	15.01	12.70	**-2.31**	**mechanical**
35	flex-top locknut	9.13	OVERSIZED washer O1.250	1.52	4.75 + 1.52 + 9.13	15.40	12.70	**-2.70**	**mechanical**
36	flex-top locknut	9.13	flat washer O0.750	1.52	4.75 + 1.52 + 9.13	15.40	12.70	**-2.70**	**mechanical**
37	HEAVY nylon-insert locknut	11.11	—	0.00	4.75 + 0.00 + 11.11	15.86	12.70	**-3.16**	**mechanical**
38	HEAVY nylon-insert locknut	11.11	OVERSIZED washer O1.250	1.52	4.75 + 1.52 + 11.11	17.39	12.70	**-4.69**	**mechanical**
39	HEAVY nylon-insert locknut	11.11	flat washer O0.750	1.52	4.75 + 1.52 + 11.11	17.39	12.70	**-4.69**	**mechanical**

SPECIFIED: JAM nut (half height) + OVERSIZED washer O1.250 + threadlocker

Of 39 permutations, 11 fit and 3 of those lock MECHANICALLY. The specified stack is a deliberate pick rather than the top of this sort: +1.66 mm of thread AND 732 mm2 of bearing - the most of anything that f
THICKNESS is what runs out in this stack, not diameter — which is why the OVERSIZED washer is no harder to fit than the standard one and was taken for free. The half-height jam nut is what pays for it.

The cost is that threadlocker is a CHEMICAL lock: it has to be cleaned off and reapplied to service the joint, where a nylon insert does not. Runner-up if you would rather have that: THIN nylon-insert locknut, n

None of this is close to a bearing problem: even the smallest annulus here is ~64x under A36/1008 mild steel's 36000 psi yield. The stack is decided by THREAD LENGTH and by locking, not by bearing.